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PROFESSIONAL SUMMARY

Senior Physics & Astronomy (double major) student with a 3.85 GPA at Faculty of Science, Cairo University. Published, as a citizen scientist, with co-authorship in a exoplanet research paper. Experienced in data analysis, using tools: Excel, Python, SQL (Microsoft SQL Server), PowerBi, and Tableau. Dedicated to advancing scientific knowledge, simplifying complex astrophysical concepts for non-specialists, and contributing to impactful research.

EDUCATION

Cairo University, Faculty of Science

Senior Physics & Astronomy Student (double major) | 2023 - 2027

- GPA: 3.855 | Grade: Excellence with honors

RESEARCH EXPERIENCES & PROJECTS

International Astronomical Search Collaboration

Citizen Scientist | Aug 2024 – Feb 2025

- Contributed to the discovery of asteroids and near-earth objects by actively participating in the IASC asteroid campaign.

NASA Exoplanet Watch Project (with PAN-AFRICA)

Participant

- Conducted data analysis on exoplanets' light curves to extract many important information about the orbital properties of the exoplanets, using Python.

Customer-Feedback Analysis

Data Analyst (Academic & Personal Project)

- Analyzed a dataset containing **1949 rows and 21 columns of customer-feedback data** using Excel's advanced tools
- Created interactive, dynamical dashboard and pivot tables to visualize key metrics, resulting in **average over all experience 3.8**.

Sales Distribution Analysis

Data Analyst (Academic & Personal Project)

- Analyzed a dataset containing **870 rows, separated into two tables, and 12 columns of sales-distribution data** using Excel's advanced tools.
- Created interactive, dynamical dashboard and pivot tables to visualize key metrics, resulting in **top 10 products sold in each country**.

Ramen-Ratings Analysis

Data Analyst (Academic & Personal Project)

- Utilized **Python (Numpy, Pandas)** to clean and preprocess a complex dataset of **2161 rows and 22 columns of ramen-ratings data**, successfully handling the missing values and data inconsistencies.
- Designed and generated compelling data visualizations using **Matplotlib and Seaborn** to translate complex numerical findings into clear, actionable insights.

Amazon Sales Analysis

Data Analyst (Academic & Personal Project)

- Analyzed a large-scale dataset of **Amazon sales transactions**, performing deep-dive analysis into revenue trends, product performance, and regional sales distribution.
- Utilized **Power Query** to clean and transform raw data, ensuring consistency across categories and handling complex data relationships for accurate reporting.
- Developed an interactive, multi-page **Power BI dashboard** using advanced **DAX measures** to visualize key metrics.

Work Experience

Fifth Physics Summer School, Faculty of Science, Cairo University | Aug 2026

Undergraduate Teaching Assistant – General Relativity | 5 Jul 2026 – 3 Sep 2026

- Instructed undergraduate students in the mathematical foundations of General Relativity, focusing on **Tensor Algebra** and **Tensor Analysis** (index notation, metric tensors, and covariant derivatives).
- Facilitated hands-on problem-solving sessions covering coordinate transformations, curvature tensors, and differential geometry basics.
- Designed and delivered sessions on **Applications of General Relativity**, guiding students through physical models, geodesic equations, and spacetime geometry.

Relaunched Academy

Astrophysics senior | Aug 2025 – Oct 2025

- Mentored and guided youth researchers and high school graduates in the field of Astrophysics.

BADU Camp, Hurghada

Astronomer

- Lead Astronomical observation sessions and explain complex celestial phenomena to camp attendees.

DEMI, iSchool

Tutor Assistant & Tutor

- Help kids, between grade 4 to grade 6, to understand the main concepts of programming and how to program and make simple, basic programs.

EXTRACURRICULAR ACTIVITIES

Cairo University Astronomy Club (CUAC)

Founder

- Established and managed the club to promote astronomy education among university students.

The Egyptian Physics Network

Member | Sep 2024 – Dec 2024

- Contributed to disseminating physics concepts and knowledge to the public

Egyptian Scouts Movement

Leader of Scouts of the Faculty of Science, Cairo University | May 2025 - present

PUBLICATIONS

- Exploring Statistical Correlations in Exoplanet Transit Parameters: A Study of Over 250 Light Curves by Citizen Scientists from Developing Countries, Journal of The American Association of Variable Star Observers.

CERTIFICATIONS

Attendance at the Fifth Physics Summer School, Department of Physics, Faculty of Science, Cairo University.

INTERNSHIPS & COURSES

Digital Egypt Pioneers Initiative – auspices of Ministry of Communications and Information Technology

Data analyst | Sep 2025 – Jun 2026

- Trained under supervising professional instructors in data analysis, and learned the tools used in data analysis for the analysis and visualization such as Excel, Python, SQL (Microsoft SQL Server), PowerBi, and Tableau.

Undergraduate Research Initiative URI

- **Research Leader Course:** Developed Skills to lead a scientific research team.
- **Keys of Scientific Research:** Trained in research methodologies, paper writing, and critical reading.

SKILLS

- **Technical Skills:** SQL Server, MySQL, Python, Microsoft Power Bi, Advanced Excel, Tableau, C++, scientific research methodologies, Data Analysis, Microsoft Power Point, Microsoft Word, Power Query.
- **Language:** Arabic (Fluent), English (B2+ Upper intermediate).